

TRACE FINDERS — SYSTEM ARCHITECTURE DOCUMENT

System Version: 10.0.0

Target Specification: SIH 2026 — Problem Statement SIH26189 (AI-Powered Criminal Network Analysis System)

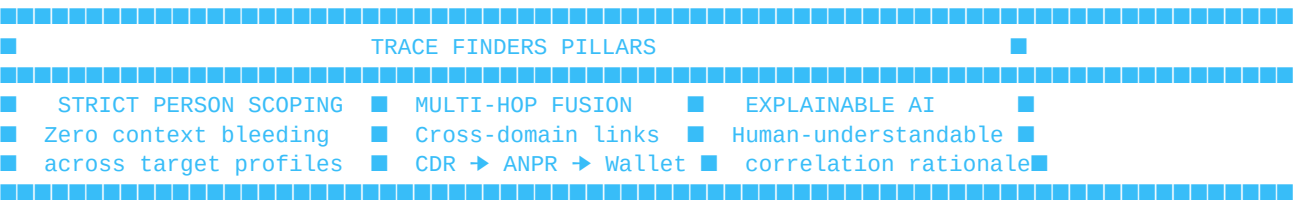
Security Level: Law Enforcement & Forensic Analyst Operational Specification

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1. Executive Summary & Architectural Philosophy

TRACE FINDERS is designed around three architectural pillars:



- 1. Strict Context Separation: The data layer enforces complete isolation between subject profiles (P-001 through P-006). No shared global references bleed across active target workstations.
- 2. Multi-Hop Evidence Fusion: Evidence records across telecom CDRs, banking ledgers, ANPR CCTV captures, and blockchain transactions are joined via graph walks.
- 3. Explainable AI (XAI): Machine-generated recommendations provide transparent reasoning, confidence metrics, and alternative hypothesis flags to satisfy court evidentiary standards.

2. End-to-End System Topology

```
flowchart TB
    subgraph Client_Layer [Client Layer]
        A[Workstation Client / Mobile Web UI]
        A1[Theme Controller: Pure OLED Pitch Black]
        A2[Context Controller: Person Switcher]
        A3[Vis.js Canvas Interactivity]
    end

    subgraph Gateway_Execution_Layer [Gateway & Execution Layer]
        B[Vercel Serverless Function Edge]
        B1[FastAPI REST API Gateway]
        B2[FastAPI Middleware & Route Handlers]
    end

    subgraph Storage_Persistence_Layer [Storage & Persistence Layer]
        C[(Firebase Realtime Database)]
        C1[Fail-Safe Local Datastore Buffer]
    end

    subgraph Analytical_Processing_Modules [Analytical Processing Modules]
        D1[Person-Scoped Graph Generator]
        D2[CCTV / DVR Forensics Engine]
        D3[Multi-Hop Fusion & XAI Engine]
        D4[A4 Dossier Report Generator]
    end

    A -->|HTTPS Requests| B
    B --> B1
    B1 --> B2
    B2 -->|Read / Write| C
    C <--> C1
    B2 --> D1
    B2 --> D2
    B2 --> D3
    B2 --> D4
```

3. Person-Scoped Data Isolation Architecture

To prevent context leakage, every entity is stored under a primary key index tied to a specific `person_id`.

```
datastore/
  ■■■ profiles/
  ■   ■■■ P-001/ (Arjun Sharma - Primary Subject)
  ■   ■■■ P-002/ (Rohan Mehta - Business Contact)
  ■   ■■■ P-003/ (Priya Joshi - Associate)
  ■   ■■■ P-004/ (Vikram Patil - Person of Interest)
  ■   ■■■ P-005/ (Neha Kulkarni - Employee)
  ■   ■■■ P-006/ (Arjun S. - Ambiguous Match Candidate)
  ■■■ cases/
  ■   ■■■ TRX-2026-017/ (Operation Nexus)
  ■■■ cctv_inventory/ (CAM-01 through CAM-12)
  ■■■ multi_hop_chains/
  ■■■ audit_logs/
```

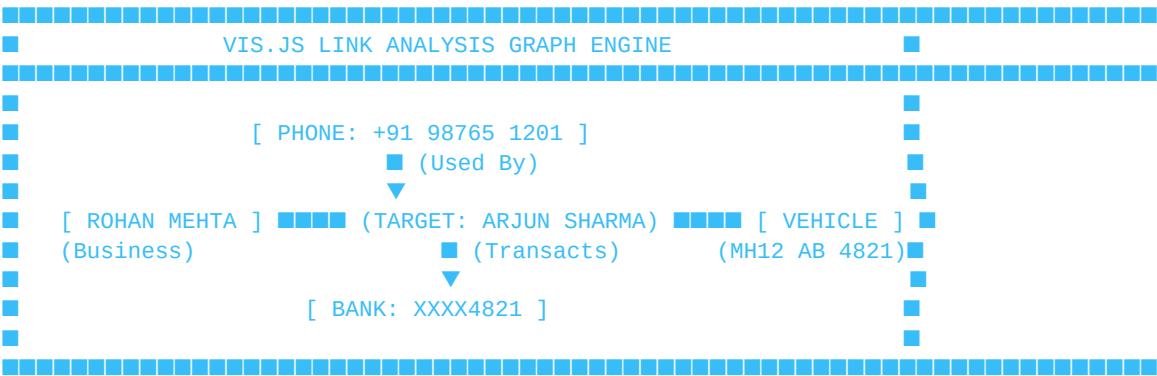
Profile Data Model Schema

```
interface PersonProfile {
  id: string;           // e.g. 'P-001'
```

```
name: string;           // e.g. 'Arjun Sharma'
alias: string;          // e.g. 'Arjun S.'
role: string;           // e.g. 'Primary Subject'
age: number;
gender: string;
phone: string;          // Isolated 10-digit primary MSISDN
email: string;          // Isolated primary email
vehicle: string;        // Isolated registration number (ANPR tag)
account_number: string; // Isolated bank ledger identifier
wallet_address: string; // Isolated EVM/UTXO blockchain wallet
photo_url: string;      // Embedded portrait URI
status: string;         // 'Under Investigation' | 'Verified'
counts: {
  calls: number;
  messages: number;
  financial: number;
  blockchain: number;
  cctv: number;
  osint: number;
};
}
```

4. Link Analysis Knowledge Graph Engine

The Link Analysis engine uses the Vis.js Network Library embedded into an unconstrained canvas container.

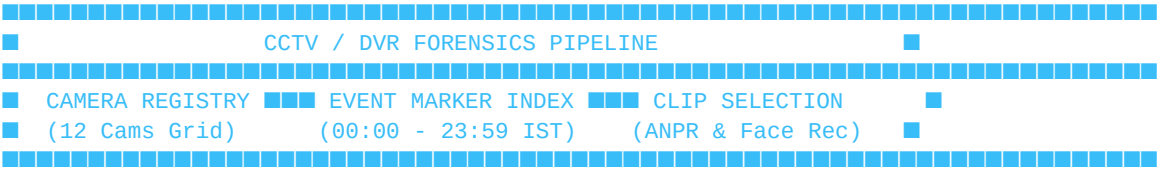


Layout Algorithms

1. Tree (Root Top): Enforces directed hierarchical level separation (UD). The currently selected target subject (person_id) is positioned dynamically as Level 0 root node.
2. Radial: Generates concentric topological distance rings radiating outward from the root subject node.
3. Hierarchy: Left-to-right (LR) chronological dependency tree.
4. Compact: Barnes-Hut gravitational force-directed clustering model ($\text{gravity} = -1000$, $\text{spring} = 50$).

5. CCTV / DVR Forensics & Timeline Pipeline

The CCTV module manages 12 active camera feeds (CAM-01 to CAM-12) without carousel truncation.

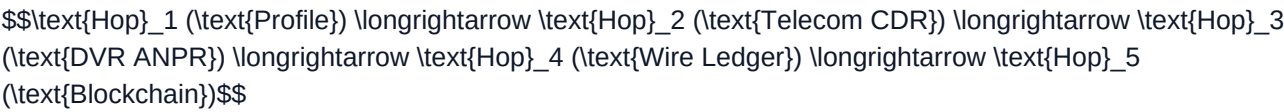


Camera Metadata Model

```
{
  "id": "CAM-04",
  "name": "Pune Central Zone",
  "location": "Shivajinagar, Pune",
  "camera_type": "Fixed CCTV",
  "status": "Active Recording",
  "resolution": "1920 x 1080",
  "recording_window": "00:00 - 23:59",
  "events_count": 12,
  "evidence_links": 8,
  "associated_persons_count": 3
}
```

6. Multi-Hop Cross-Domain Evidence Fusion & XAI Engine

The Fusion engine computes multi-hop evidence correlation paths across disparate data domains:



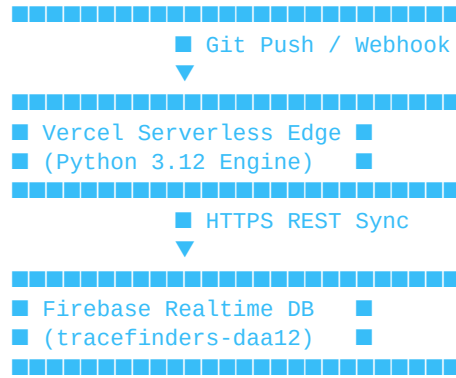
```
interface FusionChainStep {
  step: number;
  domain: 'PROFILE' | 'TELECOM' | 'CCTV' | 'FINANCIAL' | 'BLOCKCHAIN';
  label: string;
  detail: string;
  badge: string;
}

interface XAIAssessment {
  title: string;
  confidence_score: string; // e.g. "88%"
  action_status: string;    // e.g. "VERIFIED CORRELATION"
  what: string;             // Correlation rationale
  why: string;              // Flagging rationale
  alternative_explanation: string; // Counter-hypothesis for legal review
  supporting_evidence: string[]; // SHA-256 evidence record refs
}
```

7. Cloud Infrastructure & Deployment Topology

The application is deployed across a serverless hybrid cloud topology:





8. Security, Integrity & SHA-256 Chain of Custody

All evidence assets and investigative actions undergo SHA-256 cryptographic hashing prior to rendering.

```
{
  "evidence_id": "EV-COM-001",
  "case_id": "TRX-2026-017",
  "personId": "P-001",
  "acquisition_date": "18 Aug 2026 10:18 IST",
  "sha256_hash": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855",
  "custody_status": "Verified - Legal Admissible"
}
```

TRACE FINDERS Architectural Specification Document. Smart India Hackathon (SIH 2026).